

Multiple Choice

1. **Answer:** A

Options: A) a dwarf galaxy orbiting the Milky Way.; B) the closest planetary nebula to the Earth.; C) a bright star cluster discovered by Magellan.; D) the outer arm of the Milky Way named after Magellan.

Note: Correct option: A - a dwarf galaxy orbiting the Milky Way.

2. **Answer:** A

Options: A) size of the planet; B) presence of an atmosphere; C) distance from the sun; D) rotation period

Note: Correct option: A - size of the planet

3. **Answer:** A

Options: A) Its rotation is too slow.; B) It has too thick an atmosphere.; C) It is too large.; D) It does not have a metallic core.

Note: Correct option: A - Its rotation is too slow.

4. **Answer:** B

Options: A) 5; B) 7; C) 9; D) 12

Note: Correct option: B - 7

5. **Answer:** C

Options: A) Because the molecules that compose the Earth's atmosphere have a blue-ish color.; B) Because the sky reflects the color of the Earth's oceans.; C) Because the atmosphere preferentially scatters short wavelengths.; D) Because the Earth's atmosphere preferentially absorbs all other colors.

Note: Correct option: C - Because the atmosphere preferentially scatters short wavelengths.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The two stars will look like a single point of light.'

Long Form Response

11. **Answer:** Breaking of small Jovian moons to form ring materials.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** An idealized physical object that absorbs all electromagnetic radiation.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key